

The Energy Challenge in India's MSME Foundry Sector: A Problem of Scale, Cost, and Untapped Opportunity

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1 Introduction: A Sector at the Crossroads

The Kolhapur foundry cluster in Maharashtra is one of India's largest manufacturing hubs for automotive and engineering castings. Home to approximately 300 micro, small, and medium enterprises (MSMEs), the cluster produces about 600,000 tonnes of castings annually, accounting for roughly 7–8% of India's total castings production. These foundries serve critical sectors including automotive, agriculture, and heavy machinery, and employ thousands of workers across the region.

Yet beneath this industrial vitality lies a persistent and costly inefficiency: energy consumption. For most foundries, energy costs constitute between 30% and 50% of total manufacturing costs (excluding raw materials), with the melting section alone consuming nearly 70% of the total energy. In some units, the annual energy bill can exceed INR 2.5 crore (approximately USD 300,000). For a sector operating on thin margins, this is not merely an operational concern—it is an existential one.

This essay examines the energy challenge facing India's MSME foundry sector, the economic and regulatory pressures that make its resolution urgent, and the specific reasons why the Yuva Yodha Smart Manufacturing challenge provides an ideal platform for developing a solution. It argues that the sector's energy inefficiency is not primarily a technological problem—the technologies to improve efficiency exist—but rather a problem of *decision intelligence*: the absence of systematic, production-aware tools that can help SME operators make better operational decisions in real time.

2 The Scale of the Problem: Energy Consumption in Foundries

2.1 Understanding Specific Energy Consumption

The fundamental metric for energy efficiency in foundries is *Specific Energy Consumption* (SEC), measured in kilowatt-hours per tonne of metal melted or castings produced. This metric is critical because it normalises energy use against output, allowing meaningful comparisons across facilities and over time.

In Indian MSME foundries, SEC varies considerably depending on furnace type, age of equipment, and operating practices. For induction furnaces—the dominant melting

technology in many MSME units—SEC typically ranges from 600 to 770 kWh per tonne of melted metal. However, this range conceals significant variation. Two nearly identical foundries in Gujarat, India, were found to have drastically different operating costs for the same output, with SEC ranging from 625 to over 900 kWh per tonne. This variation—a spread of nearly 50% between the most and least efficient operators—represents a substantial and largely untapped opportunity for improvement.

The magnitude of this opportunity is underscored by case study data from the Kolhapur cluster. One MSME unit with an annual production of 2,672 tonnes reported an annual energy bill of INR 258 lakhs and total energy consumption of 370 tonnes of oil equivalent (toe) in the form of grid electricity. Another unit, producing 1,315 tonnes annually, reported energy costs of INR 127 lakhs, representing approximately 19% of total turnover. These figures illustrate the heavy burden that energy costs place on foundry operations.

2.2 The Sources of Inefficiency

The inefficiency in foundry energy consumption stems from multiple sources, spanning equipment, operations, and decision-making:

Equipment-level inefficiencies are the most visible. Many MSME foundries operate with aging induction furnaces, low-efficiency transformers, and poorly maintained compressors. The replacement of a 15-year-old thyristor furnace with a modern IGBT induction furnace can reduce SEC from approximately 850 kWh per tonne to 625 kWh per tonne—a saving of over 25%. Similarly, replacing low-efficiency pumps and compressors has been shown to yield payback periods of less than one year in some cases.

Operational inefficiencies are equally significant. In one documented case, a foundry’s screw air compressor was operating in unload position for 58% of the time, wasting substantial energy. In another, radiation losses from an uncovered induction furnace opening accounted for 32.7 kWh per batch—a loss that was virtually eliminated by installing a simple lid mechanism at a cost of INR 5.6 lakhs, saving 208,380 kWh annually with a payback period of just 0.4 years.

Decision-level inefficiencies are the least visible but potentially the most significant. Foundry operators typically make melting and scheduling decisions based on rules of thumb: melt as soon as possible, hold at temperature until pouring, use a fixed power level regardless of batch characteristics. These heuristics, while simple to implement, systematically waste energy through unnecessary holding time, suboptimal power trajectories, and failure to account for time-of-use electricity tariffs. The savings from better *decision-making*—not just better equipment—are substantial. A 2025 study on intelligent induction furnace control achieved a 7.6% reduction in energy consumption per metric ton of liquid metal using PLC/SCADA and LSTM-based control.

2.3 The Aggregate Impact

The cumulative effect of these inefficiencies is staggering. The Kolhapur cluster alone consumes approximately 166,910 tonnes of oil equivalent annually. At the national level, the UNIDO-BEE project working with MSME foundry clusters has documented that more than 250 units cut 18 million kWh per year—saving roughly INR 14 crore in electricity and avoiding 15,500 tonnes of CO₂ annually. Yet these achievements, while significant, represent only a fraction of the potential. A pilot study across three Kolhapur foundries

identified 11 targeted energy conservation measures that could deliver overall energy savings of up to 30%, with several measures requiring no upfront investment and delivering equipment-specific energy savings ranging from 9% to 45%.

3 Why This Problem Matters Now

3.1 The Economic Imperative

For MSME foundries, energy efficiency is not an environmental luxury—it is a competitive necessity. Energy costs typically account for 30–50% of total production costs, making them the single largest variable cost after raw materials. In some units, energy costs can reach 19% of total turnover. In a sector where profit margins are often in single digits, even a 10% reduction in energy consumption can meaningfully improve profitability.

The economic case for energy efficiency is further strengthened by the rapid payback periods documented in case studies. One Kolhapur unit achieved a 24% reduction in its total energy bill through a combination of equipment upgrades and operational improvements, with a simple payback period of 2.2 years. Another unit reduced its energy bill by 14% with a payback of just 0.7 years. A third achieved savings of INR 36 lakhs annually on an investment of INR 23 lakhs. These are not speculative returns—they are documented outcomes from real MSME foundries.

3.2 The Regulatory Pressure: CBAM

The economic imperative is now reinforced by regulatory pressure from international markets. Nearly 30% of Kolhapur’s foundry production is exported to Europe, Southeast Asia, and the Middle East. The European Union’s Carbon Border Adjustment Mechanism (CBAM), which entered full implementation in January 2026, imposes carbon costs on imported goods based on their emissions intensity.

The financial implications are substantial. India’s steel sector emits approximately 2.5 tonnes of CO₂ per tonne of steel produced—about 0.7 to 1.0 tonnes higher than the EU average. At current EU Emissions Trading System prices (approximately €79.68 per tonne of CO₂), this translates to an additional cost of €55.78 to €79.68 per tonne of steel exported (approximately INR 5,231 to INR 7,470). The impact is already visible: India’s steel and aluminium exports to the EU declined by 24.4% in FY 2025 as CBAM reporting began and most firms struggled with compliance. Industry estimates suggest that once carbon costs are fully reflected, export prices could rise by 15–22%.

For Indian foundries, CBAM is not a distant threat—it is a present reality. As one industry observer noted, “Non-EU producers like Indian foundries face duties up to €173 per tonne on steel.” The foundries that reduce their energy intensity and carbon footprint will not only avoid these penalties but may also gain a competitive advantage as global buyers increasingly prioritise low-carbon supply chains.

3.3 The Gap Between Opportunity and Action

The paradox of the Indian MSME foundry sector is that the solutions to its energy problem are largely known, yet they remain largely unimplemented. The primary reason, according to a Bureau of Energy Efficiency (BEE) assessment, is “lack of awareness.” But this diagnosis requires unpacking.

The “lack of awareness” is not simply a knowledge deficit about the existence of energy-efficient technologies. Rather, it reflects a more fundamental gap: MSME foundries lack the *decision-support tools* that would allow them to systematically identify, evaluate, and implement energy-saving opportunities in the context of their specific production constraints.

Consider the following: a foundry operator knows that reducing holding time saves energy, but does not have a tool that can calculate the optimal start time for each batch given the day’s production schedule, furnace characteristics, and electricity tariff. The operator knows that different power trajectories affect energy consumption, but does not have a model that can recommend the minimum-energy trajectory that still achieves the required metallurgical quality. The operator knows that compressed air leaks waste energy, but does not have a system that can continuously monitor and quantify these losses.

This is the critical gap. The technologies exist. The economic case is proven. What is missing is the *intelligence layer* that translates general knowledge into specific, actionable decisions for each unique production context. As the BEE assessment notes, “the main reason for the lower penetration of energy efficiency in the Kolhapur foundry sector is lack of awareness”—but this awareness gap is fundamentally a *decision-support gap*.

4 Why This Is the Right Challenge for Yuva Yodha

4.1 Alignment with the Smart Manufacturing Track

The Yuva Yodha Smart Manufacturing challenge is focused on “improving industrial energy and process efficiency while reducing emissions without affecting competitiveness.” This description captures precisely the problem facing India’s MSME foundries.

The challenge is not to reduce energy consumption at any cost—that would be trivial (simply shut down the furnace). The challenge is to reduce *specific* energy consumption (energy per unit of output) while maintaining or improving throughput, quality, and competitiveness. This is the constrained optimisation problem at the heart of industrial energy management, and it is exactly the problem that MSME foundry operators face daily.

4.2 The Fit with Schneider Electric’s Strategic Direction

Schneider Electric has demonstrated that unified power and process management can deliver significant energy savings at large industrial sites. The EcoStruxure Power and Process platform, for example, achieves real-time energy optimisation by aligning power data with production schedules. However, these solutions are designed for enterprise-scale deployments with substantial instrumentation, integration, and technical expertise.

The MSME foundry sector represents a strategic gap in Schneider’s current portfolio: a large, energy-intensive market segment that cannot afford or absorb enterprise-grade solutions but desperately needs better energy management. A solution that addresses this gap—one that is lightweight, low-cost, and requires minimal instrumentation—would not compete with Schneider’s existing products but would complement them, extending the benefits of production-energy co-optimisation to a previously underserved segment.

4.3 The Opportunity for Genuine Impact

The impact potential of a successful solution is substantial. If a production-aware energy optimisation system could capture even a fraction of the documented 9–45% energy savings in the Kolhapur cluster, the aggregate effect would be transformative. For a typical MSME foundry with an annual energy bill of INR 100 lakhs, a 15% reduction would save INR 15 lakhs per year—money that could be reinvested in equipment, workforce development, or competitive pricing.

Beyond the immediate economic benefits, such a solution would help Indian foundries maintain and expand their access to international markets by reducing their carbon footprint and CBAM exposure. It would also contribute to India’s broader sustainability goals, including the reduction of industrial emissions and the promotion of energy efficiency in the MSME sector.

4.4 The Technical and Intellectual Challenge

From a technical perspective, the problem is rich and non-trivial. The optimisation of induction furnace melting involves:

- **Thermal dynamics:** The temperature evolution of the melt depends on power input, heat losses, charge mass, and material properties—a coupled, nonlinear system.
- **Production constraints:** Each batch has a deadline, a target temperature, and quality requirements that must be met.
- **Scheduling complexity:** The order and timing of batches affect holding losses, furnace utilisation, and energy consumption.
- **Uncertainty:** Furnace efficiency varies with operating conditions, charge composition, and age; measurements are imperfect.

This combination of physics, scheduling, optimisation, and uncertainty creates a problem that is intellectually challenging yet practically important—precisely the kind of problem that a hackathon should address.

5 Conclusion: A Problem Worth Solving

The energy challenge in India’s MSME foundry sector is not a problem of technology—the technologies to improve efficiency exist and have been proven in case studies across the country. It is not a problem of economics—the return on investment for energy efficiency measures is well documented and often rapid. It is, at its core, a problem of *decision intelligence*: the absence of systematic, production-aware tools that can help SME operators make better operational decisions in real time.

The Kolhapur foundry cluster, with its 300 units, 600,000 tonnes of annual production, and documented energy-saving opportunities of 9–45%, provides a compelling context for developing such a tool. The economic pressures of high energy costs, the regulatory pressures of CBAM, and the competitive pressures of global markets make the need urgent. And the Yuva Yodha Smart Manufacturing challenge provides the ideal platform

for developing a solution that is practical, scalable, and aligned with Schneider Electric’s strategic direction.

The problem is well-defined. The opportunity is substantial. The time to act is now.

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